

ECEN 629 Applied Convex Optimization

Chapter 4. Convex optimization problems

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Optimization problem in standard form

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- $x \in \mathbf{R}^n$ is the optimization variable
- $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$ is the objective or cost function
- $f_i : \mathbf{R}^n \rightarrow \mathbf{R}$, $i = 1, \dots, m$, are the inequality constraint functions
- $h_i : \mathbf{R}^n \rightarrow \mathbf{R}$ are the equality constraint functions

optimal value:

$$p^* = \inf \{ f_0(x) \mid f_i(x) \leq 0, \ i = 1, \dots, m, \ h_i(x) = 0, \ i = 1, \dots, p \}$$

- $p^* = \infty$ if problem is infeasible (no x satisfies the constraints)
- $p^* = -\infty$ if problem is unbounded below

Optimal and locally optimal points

x is **feasible** if $x \in \text{dom } f_0$ and it satisfies the constraints

a feasible x is **optimal** if $f_0(x) = p^*$; X_{opt} is the set of optimal points

x is **locally optimal** if there is an $R > 0$ such that x is optimal for

$$\begin{array}{ll} \text{minimize (over } z) & f_0(z) \\ \text{subject to} & f_i(z) \leq 0, \quad i = 1, \dots, m, \quad h_i(z) = 0, \quad i = 1, \dots, p \\ & \|z - x\|_2 \leq R \end{array}$$

examples (with $n = 1$, $m = p = 0$)

- $f_0(x) = 1/x$, $\text{dom } f_0 = \mathbf{R}_{++}$: $p^* = 0$, no optimal point
- $f_0(x) = -\log x$, $\text{dom } f_0 = \mathbf{R}_{++}$: $p^* = -\infty$
- $f_0(x) = x \log x$, $\text{dom } f_0 = \mathbf{R}_{++}$: $p^* = -1/e$, $x = 1/e$ is optimal
- $f_0(x) = x^3 - 3x$, $p^* = -\infty$, local optimum at $x = 1$

Exercise 1

Consider the following problem:

$$\begin{aligned} & \text{minimize: } f_0(x_1, x_2) \\ & \text{subject to: } 2x_1 + x_2 \geq 1 \\ & \quad \quad \quad x_1 + 3x_2 \geq 1 \\ & \quad \quad \quad x_1 \geq 0, x_2 \geq 0 \end{aligned}$$

Solve the problem for the following objective functions, respectively:

1. $f_0(x_1, x_2) = x_1 + x_2$
2. $f_0(x_1, x_2) = x_1 - x_2$
3. $f_0(x_1, x_2) = x_1$
4. $f_0(x_1, x_2) = \max(x_1, x_2)$
5. $f_0(x_1, x_2) = x_1^2 + 9x_2^2$

Implicit constraints

the standard form optimization problem has an **implicit constraint**

$$x \in \mathcal{D} = \bigcap_{i=0}^m \text{dom } f_i \cap \bigcap_{i=1}^p \text{dom } h_i,$$

- we call $\mathcal{D}$ the **domain** of the problem
- the constraints $f_i(x) \leq 0$, $h_i(x) = 0$ are the explicit constraints
- a problem is **unconstrained** if it has no explicit constraints ($m = p = 0$)

example:

$$\text{minimize } f_0(x) = -\sum_{i=1}^k \log(b_i - a_i^T x)$$

is an unconstrained problem with implicit constraints $a_i^T x < b_i$

Feasibility problem

$$\begin{array}{ll}\text{find} & x \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

can be considered a special case of the general problem with $f_0(x) = 0$:

$$\begin{array}{ll}\text{minimize} & 0 \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

- $p^* = 0$ if constraints are feasible; any feasible x is optimal
- $p^* = \infty$ if constraints are infeasible

Convex optimization problem

standard form convex optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & a_i^T x = b_i, \quad i = 1, \dots, p\end{array}$$

- $f_0, f_1, \dots, f_m$ are convex; equality constraints are affine
- problem is *quasiconvex* if f_0 is quasiconvex (and $f_1, \dots, f_m$ convex)

often written as

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

important property: feasible set of a convex optimization problem is convex

example

$$\begin{array}{ll}\text{minimize} & f_0(x) = x_1^2 + x_2^2 \\ \text{subject to} & f_1(x) = x_1/(1 + x_2^2) \leq 0 \\ & h_1(x) = (x_1 + x_2)^2 = 0\end{array}$$

- f_0 is convex; feasible set $\{(x_1, x_2) \mid x_1 = -x_2 \leq 0\}$ is convex
- not a convex problem (according to our definition): f_1 is not convex, h_1 is not affine
- equivalent (but not identical) to the convex problem

$$\begin{array}{ll}\text{minimize} & x_1^2 + x_2^2 \\ \text{subject to} & x_1 \leq 0 \\ & x_1 + x_2 = 0\end{array}$$

Local and global optima

any locally optimal point of a convex problem is (globally) optimal

proof: suppose x is locally optimal, but there exists a feasible y with $f_0(y) < f_0(x)$

x locally optimal means there is an $R > 0$ such that

$$z \text{ feasible, } \|z - x\|_2 \leq R \implies f_0(z) \geq f_0(x)$$

consider $z = \theta y + (1 - \theta)x$ with $\theta = R/(2\|y - x\|_2)$

- $\|y - x\|_2 > R$, so $0 < \theta < 1/2$
- z is a convex combination of two feasible points, hence also feasible
- $\|z - x\|_2 = R/2$ and

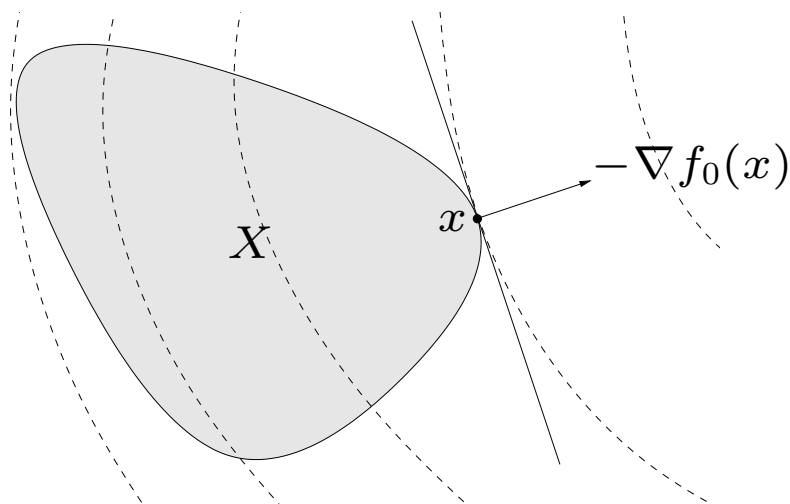
$$f_0(z) \leq \theta f_0(y) + (1 - \theta)f_0(x) < f_0(x)$$

which contradicts our assumption that x is locally optimal

Optimality criterion for differentiable f_0

x is optimal if and only if it is feasible and

$$\nabla f_0(x)^T(y - x) \geq 0 \quad \text{for all feasible } y$$



if nonzero, $\nabla f_0(x)$ defines a supporting hyperplane to feasible set X at x

- **unconstrained problem:** x is optimal if and only if

$$x \in \mathbf{dom} f_0, \quad \nabla f_0(x) = 0$$

- **equality constrained problem**

$$\text{minimize } f_0(x) \quad \text{subject to } Ax = b$$

x is optimal if and only if there exists a ν such that

$$x \in \mathbf{dom} f_0, \quad Ax = b, \quad \nabla f_0(x) + A^T \nu = 0$$

- **minimization over nonnegative orthant**

$$\text{minimize } f_0(x) \quad \text{subject to } x \succeq 0$$

x is optimal if and only if

$$x \in \mathbf{dom} f_0, \quad x \succeq 0, \quad \begin{cases} \nabla f_0(x)_i \geq 0 & x_i = 0 \\ \nabla f_0(x)_i = 0 & x_i > 0 \end{cases}$$

Equivalent convex problems

two problems are (informally) **equivalent** if the solution of one is readily obtained from the solution of the other, and vice-versa

some common transformations that preserve convexity:

- **eliminating equality constraints**

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize (over } z) & f_0(Fz + x_0) \\ \text{subject to} & f_i(Fz + x_0) \leq 0, \quad i = 1, \dots, m\end{array}$$

where F and x_0 are such that

$$Ax = b \iff x = Fz + x_0 \text{ for some } z$$

- **introducing equality constraints**

$$\begin{array}{ll}\text{minimize} & f_0(A_0x + b_0) \\ \text{subject to} & f_i(A_ix + b_i) \leq 0, \quad i = 1, \dots, m\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize (over } x, y_i) & f_0(y_0) \\ \text{subject to} & f_i(y_i) \leq 0, \quad i = 1, \dots, m \\ & y_i = A_ix + b_i, \quad i = 0, 1, \dots, m\end{array}$$

- **introducing slack variables for linear inequalities**

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

is equivalent to

$$\begin{array}{ll}\text{minimize (over } x, s) & f_0(x) \\ \text{subject to} & a_i^T x + s_i = b_i, \quad i = 1, \dots, m \\ & s_i \geq 0, \quad i = 1, \dots, m\end{array}$$

- **epigraph form:** standard form convex problem is equivalent to

$$\begin{array}{ll} \text{minimize (over } x, t) & t \\ \text{subject to} & f_0(x) - t \leq 0 \\ & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b \end{array}$$

- **minimizing over some variables**

$$\begin{array}{ll} \text{minimize} & f_0(x_1, x_2) \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \dots, m \end{array}$$

is equivalent to

$$\begin{array}{ll} \text{minimize} & \tilde{f}_0(x_1) \\ \text{subject to} & f_i(x_1) \leq 0, \quad i = 1, \dots, m \end{array}$$

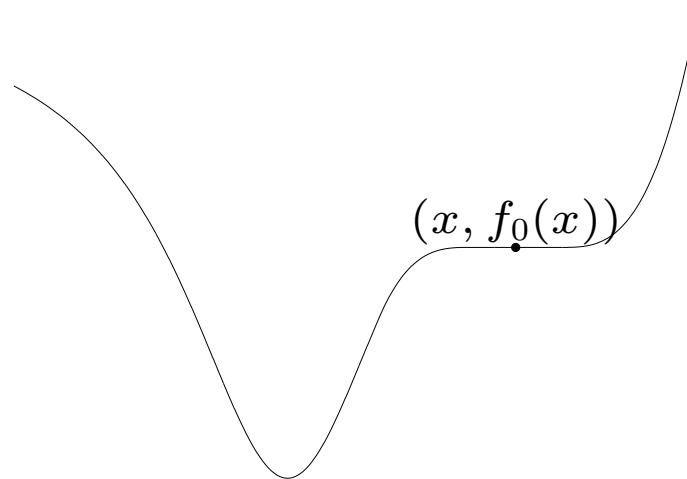
where $\tilde{f}_0(x_1) = \inf_{x_2} f_0(x_1, x_2)$

Quasiconvex optimization

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

with $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$ quasiconvex, $f_1, \dots, f_m$ convex

can have locally optimal points that are not (globally) optimal



convex representation of sublevel sets of f_0

if f_0 is quasiconvex, there exists a family of functions ϕ_t such that:

- $\phi_t(x)$ is convex in x for fixed t
- t -sublevel set of f_0 is 0-sublevel set of ϕ_t , i.e.,

$$f_0(x) \leq t \iff \phi_t(x) \leq 0$$

example

$$f_0(x) = \frac{p(x)}{q(x)}$$

with p convex, q concave, and $p(x) \geq 0$, $q(x) > 0$ on $\text{dom } f_0$

can take $\phi_t(x) = p(x) - tq(x)$:

- for $t \geq 0$, ϕ_t convex in x
- $p(x)/q(x) \leq t$ if and only if $\phi_t(x) \leq 0$

quasiconvex optimization via convex feasibility problems

$$\phi_t(x) \leq 0, \quad f_i(x) \leq 0, \quad i = 1, \dots, m, \quad Ax = b \quad (1)$$

- for fixed t , a convex feasibility problem in x
- if feasible, we can conclude that $t \geq p^*$; if infeasible, $t \leq p^*$

Bisection method for quasiconvex optimization

given $l \leq p^*$, $u \geq p^*$, tolerance $\epsilon > 0$.

repeat

1. $t := (l + u)/2$.
2. Solve the convex feasibility problem (1).
3. **if** (1) is feasible, $u := t$; **else** $l := t$.

until $u - l \leq \epsilon$.

requires exactly $\lceil \log_2((u - l)/\epsilon) \rceil$ iterations (where u, l are initial values)

Concave Maximization Problems

The optimization problem

$$\begin{aligned} & \text{maximize: } f_0(x) \\ & \text{subject to: } f_i(x) \leq 0, \quad i = 1, 2, \dots, m \\ & \quad Ax = b. \end{aligned}$$

is a convex optimization problem if

- f_0 is concave,
- $f_i, i = 1, 2, \dots, m$ are convex,

since it is equivalent to minimizing $-f_0$.

Abstract Form Convex Optimization Problem

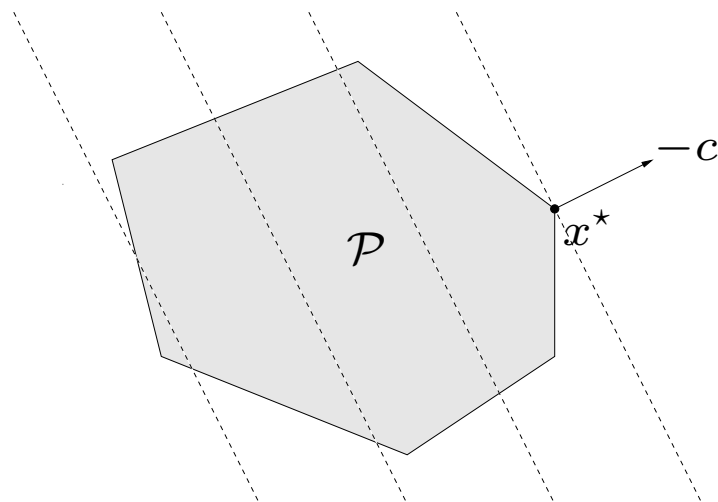
Minimizing a convex function over a convex set

- Sometimes referred to as an abstract convex optimization problem;
- The definition of “convex optimization problem” in the textbook is more strict:
 - In principle both problems have the same difficulty;
 - In practice they are the same, once the constraint set is properly represented.

Linear program (LP)

$$\begin{array}{ll}\text{minimize} & c^T x + d \\ \text{subject to} & Gx \preceq h \\ & Ax = b\end{array}$$

- convex problem with affine objective and constraint functions
- feasible set is a polyhedron



Standard Form Linear Programs

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Ax = b \\ &\quad x \succeq 0. \end{aligned}$$

All LPs can be converted to standard form LPs: e.g., the form on the last page

$$\begin{aligned} &\text{minimize: } c^T x + d \\ &\text{subject to: } Gx \preceq h \\ &\quad Ax = b. \end{aligned}$$

- Introduce slack variables s_i , where $s \succeq 0$: $Gx + s = h$
- Express x as $x = x^+ - x^-$, where $x^+, x^- \succeq 0$:

$$\begin{aligned} &\text{minimize: } c^T x^+ - c^T x^- + d \\ &\text{subject to: } Gx^+ - Gx^- + s = h \\ &\quad Ax = b \\ &\quad x^+ \succeq 0, x^- \succeq 0, s \succeq 0. \end{aligned}$$

Exercise 2

Consider the LP

$$\begin{aligned} &\text{minimize: } c^T x \\ &\text{subject to: } Ax \preceq b, \end{aligned}$$

where A is $n \times n$ and nonsingular (invertible).

- What is the optimal value and optimal set?

Examples

diet problem: choose quantities $x_1, \dots, x_n$ of n foods

- one unit of food j costs c_j , contains amount a_{ij} of nutrient i
- healthy diet requires nutrient i in quantity at least b_i

to find cheapest healthy diet,

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \succeq b, \quad x \succeq 0\end{array}$$

piecewise-linear minimization

$$\text{minimize} \quad \max_{i=1,\dots,m} (a_i^T x + b_i)$$

equivalent to an LP

$$\begin{array}{ll}\text{minimize} & t \\ \text{subject to} & a_i^T x + b_i \leq t, \quad i = 1, \dots, m\end{array}$$

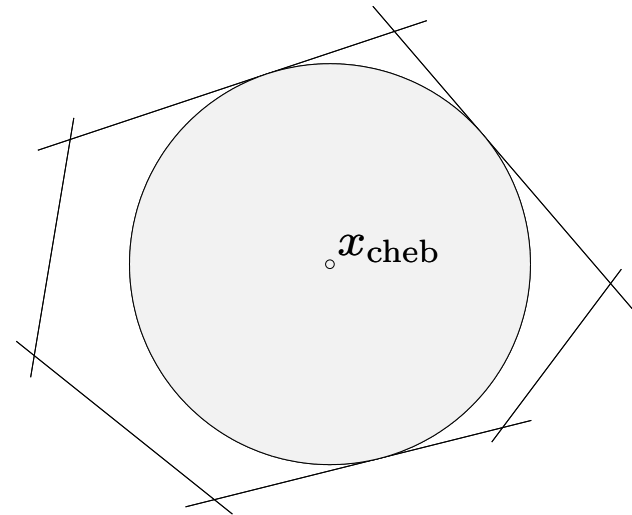
Chebyshev center of a polyhedron

Chebyshev center of

$$\mathcal{P} = \{x \mid a_i^T x \leq b_i, \ i = 1, \dots, m\}$$

is center of largest inscribed ball

$$\mathcal{B} = \{x_c + u \mid \|u\|_2 \leq r\}$$



- $a_i^T x \leq b_i$ for all $x \in \mathcal{B}$ if and only if

$$\sup\{a_i^T (x_c + u) \mid \|u\|_2 \leq r\} = a_i^T x_c + r\|a_i\|_2 \leq b_i$$

- hence, x_c, r can be determined by solving the LP

$$\begin{array}{ll} \text{maximize} & r \\ \text{subject to} & a_i^T x_c + r\|a_i\|_2 \leq b_i, \quad i = 1, \dots, m \end{array}$$

Linear-fractional program

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & Gx \preceq h \\ & Ax = b\end{array}$$

linear-fractional program

$$f_0(x) = \frac{c^T x + d}{e^T x + f}, \quad \text{dom } f_0(x) = \{x \mid e^T x + f > 0\}$$

- a quasiconvex optimization problem; can be solved by bisection
- also equivalent to the LP (variables y, z)

$$\begin{array}{ll}\text{minimize} & c^T y + dz \\ \text{subject to} & Gy \preceq hz \\ & Ay = bz \\ & e^T y + fz = 1 \\ & z \geq 0\end{array}$$

Transforming Linear-fractional Programming

$$\begin{aligned} &\text{minimize: } \frac{c^T x + d}{e^T x + f} \\ &\text{subject to: } Gx \preceq h \\ &\quad Ax = b, \end{aligned}$$

where $e^T x + f > 0$.

$$\begin{aligned} &\text{minimize: } c^T y + dz \\ &\text{subject to: } Gy - hz \preceq 0 \\ &\quad Ay - bz = 0 \\ &\quad e^T y + fz = 1 \\ &\quad z \geq 0. \end{aligned}$$

Transforming Linear-fractional Programming

$$\begin{aligned} \text{minimize: } & \frac{c^T x + d}{e^T x + f} \\ \text{subject to: } & Gx \preceq h \\ & Ax = b, \end{aligned}$$

$$\begin{aligned} \text{minimize: } & c^T y + dz \\ \text{subject to: } & Gy - hz \preceq 0 \\ & Ay - bz = 0 \\ & e^T y + fz = 1 \\ & z \geq 0. \end{aligned}$$

where $e^T x + f > 0$.

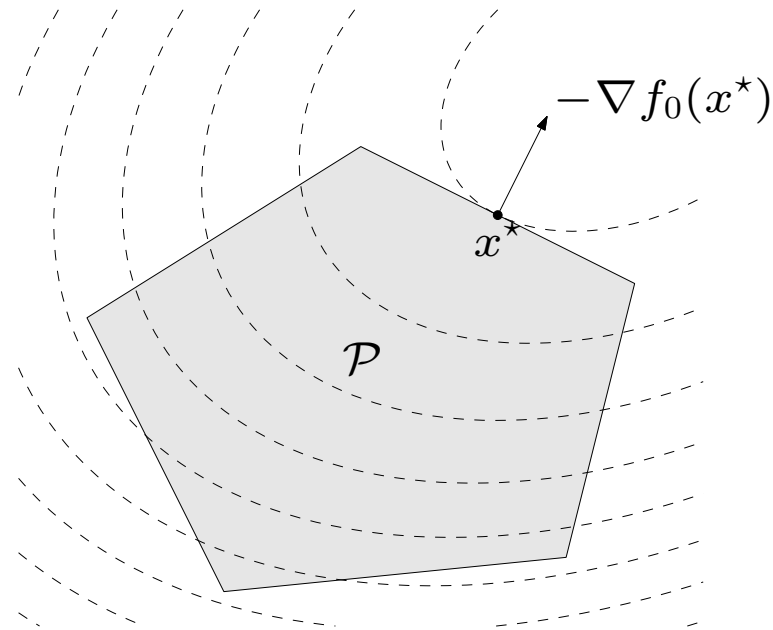
- $(P^*) \geq (P_{y,z}^*)$: $y = \frac{x}{e^T x + f}$, $z = \frac{1}{e^T x + f}$.
- $(P^*) \leq (P_{y,z}^*)$?: consider feasible (y, z)
 - If $z \neq 0$: then $x = y/z$.
 - If $z = 0$: $x = x_0 + ty$ is feasible for LHS with x_0 feasible in LHS and y on RHS for any $t \geq 0$

$$\lim_{t \rightarrow \infty} \frac{c^T(x_0 + ty) + d}{e^T(x_0 + ty) + f} = \frac{c^T y}{e^T y} = c^T y = c^T y + dz.$$

Quadratic program (QP)

$$\begin{array}{ll}\text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Gx \preceq h \\ & Ax = b\end{array}$$

- $P \in \mathbf{S}_{+}^n$, so objective is convex quadratic
- minimize a convex quadratic function over a polyhedron



Examples

least-squares

$$\text{minimize} \quad \|Ax - b\|_2^2$$

- analytical solution $x^* = A^\dagger b$ ($A^\dagger$ is pseudo-inverse)
- can add linear constraints, *e.g.*, $l \preceq x \preceq u$

linear program with random cost

$$\begin{array}{ll} \text{minimize} & \bar{c}^T x + \gamma x^T \Sigma x = \mathbf{E} c^T x + \gamma \mathbf{var}(c^T x) \\ \text{subject to} & Gx \preceq h, \quad Ax = b \end{array}$$

- c is random vector with mean $\bar{c}$ and covariance Σ
- hence, $c^T x$ is random variable with mean $\bar{c}^T x$ and variance $x^T \Sigma x$
- $\gamma > 0$ is risk aversion parameter; controls the trade-off between expected cost and variance (risk)

Quadratically constrained quadratic program (QCQP)

$$\begin{array}{ll}\text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

- $P_i \in \mathbf{S}_{+}^n$; objective and constraints are convex quadratic
- if $P_1, \dots, P_m \in \mathbf{S}_{++}^n$, feasible region is intersection of m ellipsoids and an affine set

Second-order cone programming

$$\begin{array}{ll}\text{minimize} & f^T x \\ \text{subject to} & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \dots, m \\ & Fx = g\end{array}$$

$$(A_i \in \mathbf{R}^{n_i \times n}, F \in \mathbf{R}^{p \times n})$$

- inequalities are called second-order cone (SOC) constraints:

$$(A_i x + b_i, c_i^T x + d_i) \in \text{second-order cone in } \mathbf{R}^{n_i+1}$$

- for $n_i = 0$, reduces to an LP; if $c_i = 0$, reduces to a QCQP
- more general than QCQP and LP

Robust linear programming

the parameters in optimization problems are often uncertain, *e.g.*, in an LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m,\end{array}$$

there can be uncertainty in c , a_i , b_i

two common approaches to handling uncertainty (in a_i , for simplicity)

- deterministic model: constraints must hold for all $a_i \in \mathcal{E}_i$

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i \text{ for all } a_i \in \mathcal{E}_i, \quad i = 1, \dots, m,\end{array}$$

- stochastic model: a_i is random variable; constraints must hold with probability η

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & \mathbf{prob}(a_i^T x \leq b_i) \geq \eta, \quad i = 1, \dots, m\end{array}$$

deterministic approach via SOCP

- choose an ellipsoid as $\mathcal{E}_i$:

$$\mathcal{E}_i = \{\bar{a}_i + P_i u \mid \|u\|_2 \leq 1\} \quad (\bar{a}_i \in \mathbf{R}^n, \quad P_i \in \mathbf{R}^{n \times n})$$

center is $\bar{a}_i$, semi-axes determined by singular values/vectors of P_i

- robust LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i \quad \forall a_i \in \mathcal{E}_i, \quad i = 1, \dots, m \end{array}$$

is equivalent to the SOCP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \|P_i^T x\|_2 \leq b_i, \quad i = 1, \dots, m \end{array}$$

(follows from $\sup_{\|u\|_2 \leq 1} (\bar{a}_i + P_i u)^T x = \bar{a}_i^T x + \|P_i^T x\|_2$)

stochastic approach via SOCP

- assume a_i is Gaussian with mean $\bar{a}_i$, covariance Σ_i ($a_i \sim \mathcal{N}(\bar{a}_i, \Sigma_i)$)
- $a_i^T x$ is Gaussian r.v. with mean $\bar{a}_i^T x$, variance $x^T \Sigma_i x$; hence

$$\mathbf{prob}(a_i^T x \leq b_i) = \Phi \left(\frac{b_i - \bar{a}_i^T x}{\|\Sigma_i^{1/2} x\|_2} \right)$$

where $\Phi(x) = (1/\sqrt{2\pi}) \int_{-\infty}^x e^{-t^2/2} dt$ is CDF of $\mathcal{N}(0, 1)$

- robust LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \mathbf{prob}(a_i^T x \leq b_i) \geq \eta, \quad i = 1, \dots, m, \end{array}$$

with $\eta \geq 1/2$, is equivalent to the SOCP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \Phi^{-1}(\eta) \|\Sigma_i^{1/2} x\|_2 \leq b_i, \quad i = 1, \dots, m \end{array}$$

Geometric programming

monomial function

$$f(x) = cx_1^{a_1}x_2^{a_2}\cdots x_n^{a_n}, \quad \text{dom } f = \mathbf{R}_{++}^n$$

with $c > 0$; exponent a_i can be any real number

posynomial function: sum of monomials

$$f(x) = \sum_{k=1}^K c_k x_1^{a_{1k}} x_2^{a_{2k}} \cdots x_n^{a_{nk}}, \quad \text{dom } f = \mathbf{R}_{++}^n$$

geometric program (GP)

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 1, \quad i = 1, \dots, m \\ & h_i(x) = 1, \quad i = 1, \dots, p \end{array}$$

with f_i posynomial, h_i monomial

Geometric program in convex form

change variables to $y_i = \log x_i$, and take logarithm of cost, constraints

- monomial $f(x) = cx_1^{a_1} \cdots x_n^{a_n}$ transforms to

$$\log f(e^{y_1}, \dots, e^{y_n}) = a^T y + b \quad (b = \log c)$$

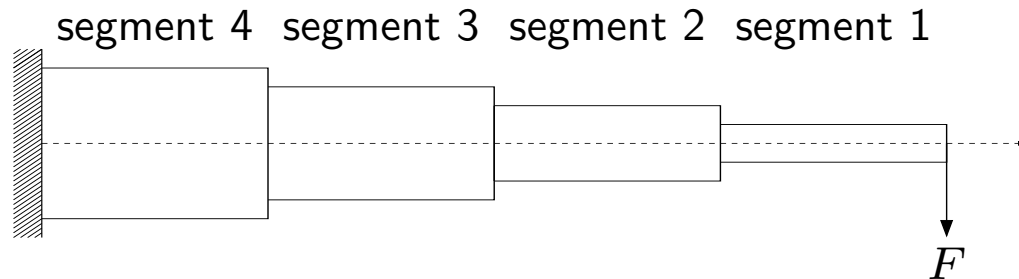
- posynomial $f(x) = \sum_{k=1}^K c_k x_1^{a_{1k}} x_2^{a_{2k}} \cdots x_n^{a_{nk}}$ transforms to

$$\log f(e^{y_1}, \dots, e^{y_n}) = \log \left(\sum_{k=1}^K e^{a_k^T y + b_k} \right) \quad (b_k = \log c_k)$$

- geometric program transforms to convex problem

$$\begin{array}{ll} \text{minimize} & \log \left(\sum_{k=1}^K \exp(a_{0k}^T y + b_{0k}) \right) \\ \text{subject to} & \log \left(\sum_{k=1}^K \exp(a_{ik}^T y + b_{ik}) \right) \leq 0, \quad i = 1, \dots, m \\ & Gy + d = 0 \end{array}$$

Design of cantilever beam



- N segments with unit lengths, rectangular cross-sections of size $w_i \times h_i$
- given vertical force F applied at the right end

design problem

minimize total weight

subject to upper & lower bounds on w_i, h_i

upper bound & lower bounds on aspect ratios h_i/w_i

upper bound on stress in each segment

upper bound on vertical deflection at the end of the beam

variables: w_i, h_i for $i = 1, \dots, N$

objective and constraint functions

- total weight $w_1h_1 + \cdots + w_Nh_N$ is posynomial
- aspect ratio h_i/w_i and inverse aspect ratio w_i/h_i are monomials
- maximum stress in segment i is given by $6iF/(w_ih_i^2)$, a monomial
- the vertical deflection y_i and slope v_i of central axis at the right end of segment i are defined recursively as

$$v_i = 12(i - 1/2) \frac{F}{Ew_ih_i^3} + v_{i+1}$$
$$y_i = 6(i - 1/3) \frac{F}{Ew_ih_i^3} + v_{i+1} + y_{i+1}$$

for $i = N, N - 1, \dots, 1$, with $v_{N+1} = y_{N+1} = 0$ (E is Young's modulus)

v_i and y_i are posynomial functions of w, h

formulation as a GP

$$\begin{aligned} \text{minimize} \quad & w_1 h_1 + \cdots + w_N h_N \\ \text{subject to} \quad & w_{\max}^{-1} w_i \leq 1, \quad w_{\min} w_i^{-1} \leq 1, \quad i = 1, \dots, N \\ & h_{\max}^{-1} h_i \leq 1, \quad h_{\min} h_i^{-1} \leq 1, \quad i = 1, \dots, N \\ & S_{\max}^{-1} w_i^{-1} h_i \leq 1, \quad S_{\min} w_i h_i^{-1} \leq 1, \quad i = 1, \dots, N \\ & 6iF\sigma_{\max}^{-1} w_i^{-1} h_i^{-2} \leq 1, \quad i = 1, \dots, N \\ & y_{\max}^{-1} y_1 \leq 1 \end{aligned}$$

note

- we write $w_{\min} \leq w_i \leq w_{\max}$ and $h_{\min} \leq h_i \leq h_{\max}$

$$w_{\min}/w_i \leq 1, \quad w_i/w_{\max} \leq 1, \quad h_{\min}/h_i \leq 1, \quad h_i/h_{\max} \leq 1$$

- we write $S_{\min} \leq h_i/w_i \leq S_{\max}$ as

$$S_{\min} w_i / h_i \leq 1, \quad h_i / (w_i S_{\max}) \leq 1$$

Minimizing spectral radius of nonnegative matrix

Perron-Frobenius eigenvalue $\lambda_{\text{pf}}(A)$

- exists for (elementwise) positive $A \in \mathbf{R}^{n \times n}$
- a real, positive eigenvalue of A , equal to spectral radius $\max_i |\lambda_i(A)|$
- determines asymptotic growth (decay) rate of A^k : $A^k \sim \lambda_{\text{pf}}^k$ as $k \rightarrow \infty$
- alternative characterization: $\lambda_{\text{pf}}(A) = \inf\{\lambda \mid Av \preceq \lambda v \text{ for some } v \succ 0\}$

minimizing spectral radius of matrix of posynomials

- minimize $\lambda_{\text{pf}}(A(x))$, where the elements $A(x)_{ij}$ are posynomials of x
- equivalent geometric program:

$$\begin{array}{ll} \text{minimize} & \lambda \\ \text{subject to} & \sum_{j=1}^n A(x)_{ij} v_j / (\lambda v_i) \leq 1, \quad i = 1, \dots, n \end{array}$$

variables λ, v, x

Population of Bacterium

Time index $t = 0, 1, 2, \dots$

- Vector $p(t) \in \mathbf{R}_+^4$: population age distribution at time t ;
- i -th coordinate $p_i(t)$: total population between $i - 1$ and i hours old;
- No bacteria live more than 4 hours;
- Population propagates in time as $p(t + 1) = Ap(t)$

$$A = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ s_1 & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 \\ 0 & 0 & s_3 & 0 \end{bmatrix}$$

b_i 's are the birth rates; s_i 's are the survival rates.

Population of Bacterium

- Population propagates in time as $p(t+1) = Ap(t)$

$$A = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ s_1 & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 \\ 0 & 0 & s_3 & 0 \end{bmatrix}$$

- Consider two factors that determine the birth and survival rates:

$$b_i = b_i^{nom} (c_1/c_1^{nom})^{\alpha_i} (c_2/c_2^{nom})^{\beta_i}, \quad i = 1, 2, 3, 4$$

$$s_i = s_i^{nom} (c_1/c_1^{nom})^{\gamma_i} (c_2/c_2^{nom})^{\delta_i}, \quad i = 1, 2, 3.$$

- The general optimization problem:

maximize: λ

$$\text{subject to: } \sum_{j=1}^n A_{ij} v_j / (\lambda v_j) \leq 1, \quad i = 1, 2, \dots, n$$

$$f_i(x) \leq 1, \quad i = 1, 2, \dots, p.$$

where $f_i(x)$ is an optional posynomial constraint.

Exercise 3

Norm optimization

$$\text{minimize: } \|Ax - b\|_p$$

where $A \in \mathbf{R}^{m \times n}$ and $b \in \mathbf{R}^m$ are fixed. Find the equivalent optimization problems (that are easier to compute).

- $p = \infty$ (hint: $\|x\|_\infty = \max_i |x_i|$);
- $p = 1$.

Consider the special case $A = (1, 1)^T$, and $b = (1, 4)^T$. What are the solutions?

Exercise 4

Consider the problem

$$\begin{aligned} & \text{minimize: } c^T x \\ & \text{subject to: } x^T A x \leq 1, \end{aligned}$$

where $A \in \mathbf{S}_{++}^n$.

- Find the optimal solution for this problem;
- If $A \in \mathbf{S}^n$ but not positive semidefinite, what is the solution?

Exercise 5

Let $p(x)$ and $q(x)$ be two posynomials. Consider the optimization problems:

1.

$$\text{minimize: } \max(p(x), q(x)).$$

2.

$$\text{minimize: } \exp(p(x)) + \exp(q(x)).$$

- Is it a convex optimization problem?
- Rewrite it in a form that is easier to compute.